

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

ANSWERS

BIOLOGY

Paper 2 STRUCTURED QUESTIONS

9744/02

20th AUGUST 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [10]	
2 [11]	
3 [11]	
4 [10]	
5 [10]	
6 [8]	
7 [10]	
8 [10]	
9 [10]	
TOTAL P2 [30%]	100

This document consists of **22** printed pages and **0** blank pages.

[Turn over

Answer **all** questions.

- 1 Fig. 1.1 is an electron micrograph of structure A present in a mature organelle of a plant cell.

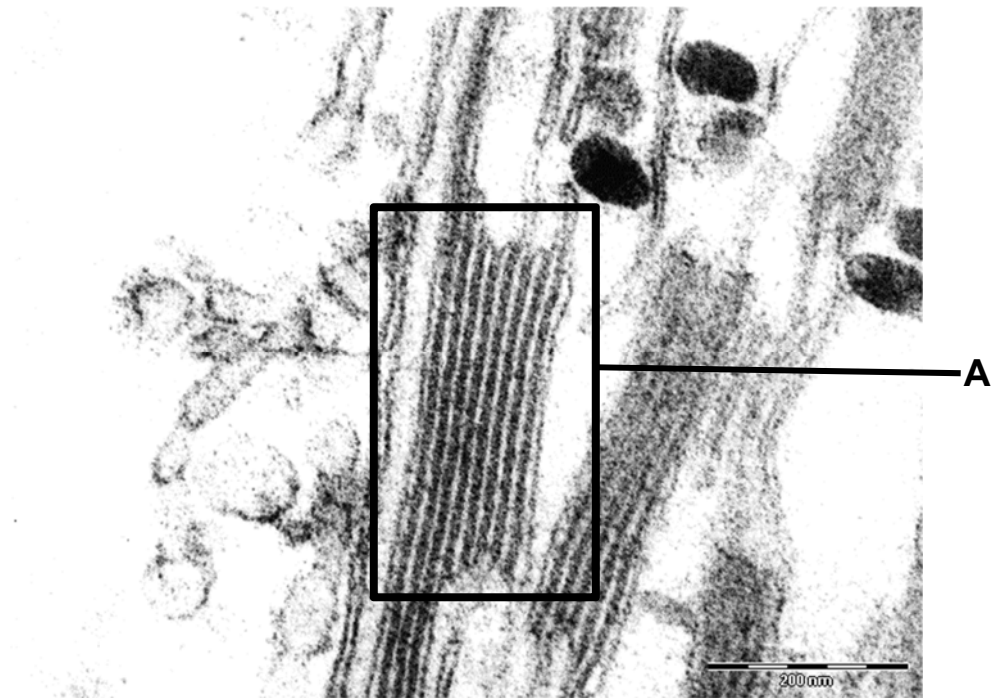


Fig. 1.1

- (a)(i) Identify the organelle in which structure A resides.

ANS [L1] Novel

SC: Identify organelle cell structure A
OR: Chloroplast

[1]

1. Chloroplast

- (a)(ii) Explain the significance of the flattened stacks arranged in structure A..

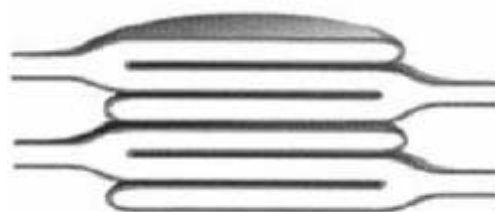
ANS [L2] Novel

SC: explain significance flattened stacks A
OR: Give reasons importance 1] increase SA; maximise attachment of photosynthetic pigments max light absorption
2] increase thylakoid space ● build up proton grad

[2]

The thylakoids are arranged in flattened sacs in stacks

1. is to provide **large surface area** to maximize the **attachment of** (photosynthetic elements) / electron protein carriers, ATP synthases and **photosynthetic pigments** for **maximum light absorption**;
2. is to **increase** thylakoid **space** for the **building up of proton gradient** needed for ATP synthesis via chemiosmosis;



Arvidsson & Sundby, 1999

(b) Fig 1.2 shows a cell fractionation experiment performed by a student on a sample of plant cells.

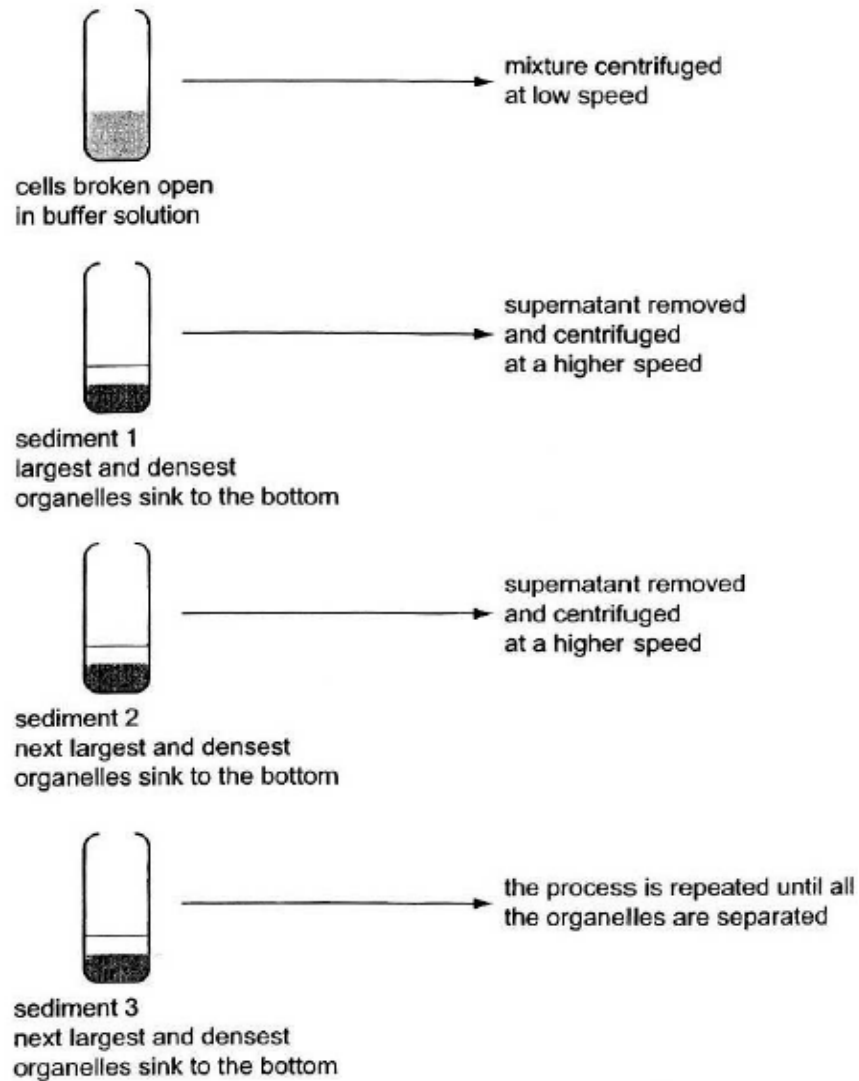


Fig. 1.2

However, she was not sure which sediment samples contained the organelles where structure A are located. Hence, she repeated the cell fractionation experiment again for all 3 sediment samples she collected, this time with a higher centrifugal speed.

Explain why there is a need for additional cell fractionation at a higher centrifugal speed.

ANS [L2] Novel

SC: Explain why

additional cell fractionation

higher centrifugal speed

OR:

mito + chloroplast (need to separate)

mature chloroplast is bigger

[2]

1. Sediment 2 contains both **mitochondria** and mature **chloroplast**
2. With additional centrifugation, mature chloroplasts which is **bigger** in size, can be separated from other smaller mitochondria / OWTTE

Fig. 1.3 shows a structure of a cell surface membrane.

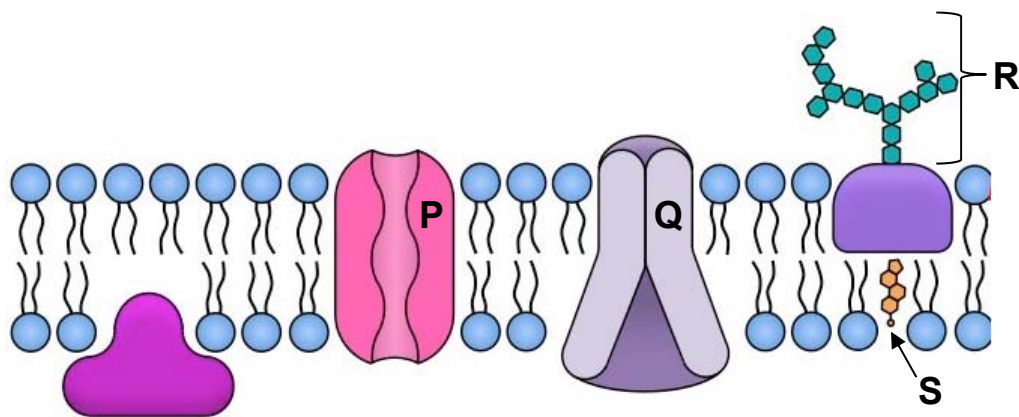


Fig. 1.3

With reference to Fig. 1.3,

(c)(i) Explain how the structure of Q allows it to be embedded in the cell surface membrane.

ANS [L1] Novel

[1]

SC: Explain how structure Q embedded
OR: made up of both hydrophilic and hydrophobic R-groups A.A

1. Structure Q / Carrier protein Q / Q has 3D conformation shape that are made up of **both hydrophilic and hydrophobic amino acids** that folds and anchored on the cell membrane where it can form hydrophilic and hydrophobic interactions with the phospholipid bilayer respectively / OWTTE

(c)(ii) Explain how P functions differently compared to Q.

ANS [L2] Novel

[1]

SC: explain how P (channel) Q (carrier) function differently
OR:

1. P allows substances to pass through cell membrane via **facilitated diffusion** only whereas Q allows both **facilitated diffusion and active transport**.

(c)(iii) Explain why a change in R may cause an immunological response.

ANS [L3] Novel

[2]

SC: suggest why mutation in R immunological response
OR: R: R ● function for cell recognition
Mutated glycoprotein R ● changes on CBH chains ● cannot recognise as self / see as foreign bodies ● trigger immune response

1. **Glycoprotein R** plays an important role for **cell-to-cell recognition** / **serve as identification tags**
2. With **mutated glycoprotein R**, this **changes** the types of sugar residues that make up the **carbohydrate chains** on glycoprotein R which is **considered foreign** / **cannot recognise as self** thus triggering an immune response.

(c)(iv) explain why structure S is amphipathic.

ANS [L2] Novel

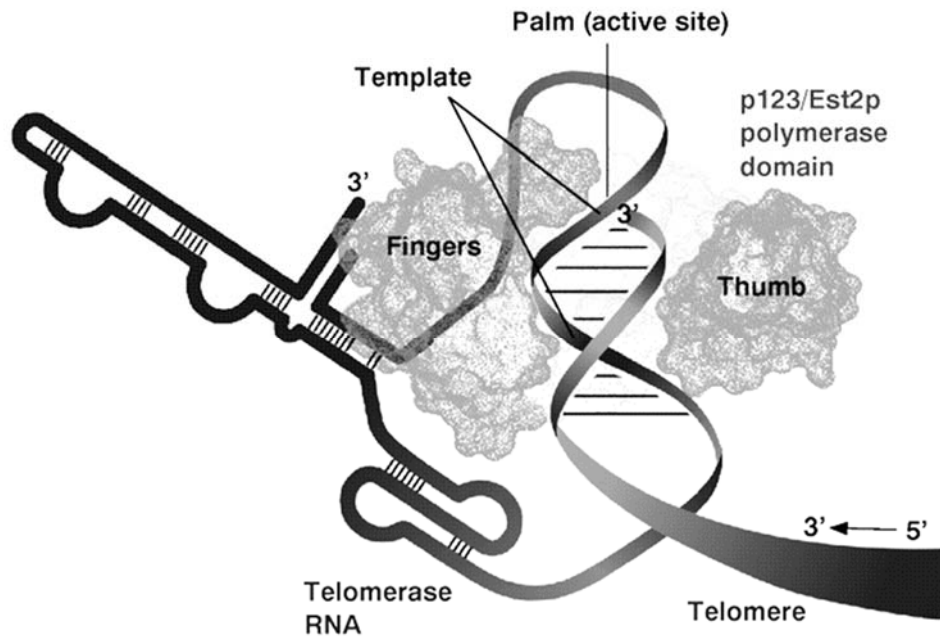
[1]

SC: Explain why structure S amphipathic
OR: S: cholesterol: 4 fused rings + hydrocarbon tails ● hydrophobic; -OH group ● hydrophilic

1. **S / cholesterol** contains both a **hydrophilic -OH group** and a **hydrophobic 4 fused C rings** that wedged neatly between the phospholipid bilayer.

- 2 Telomerase is an important enzyme that lengthens telomeres and is extensively studied for its role in cancer. Molecularly, telomerase is a unique reverse transcriptase enzyme that carries its own RNA molecule.

Fig. 2.1
commonly
model of
with its
and a
telomeric



shows a
used "hand"
the telomerase
RNA template
section of
DNA bound.

Source: <http://science.sciencemag.org/content/276/5312/561.full>

Fig. 2.1

- (a) Telomerase is hypothesised to follow the induce-fit model in binding to DNA to carry out its function of repeatedly lengthening telomeric DNA. With reference to Fig 2.1, explain what is meant by the induce-fit model.

ANS [L2] (H2 Novel)

SC: With reference
Based on information from fig
OR: RNA template recognise
Telomere 3' end is substrate

explain what is meant by induced-fit model
define the model
enzyme will change conformation upon binding to substrate
allows for free deoxyribonucleoside triphosphates to come in and form bond

[2]

1. Induced fit model refers to the **enzyme** (accept: telomerase) **undergoing conformation change slightly upon binding to its substrate**, the telomeric DNA at the 3' end;
2. For telomerase, the **substrate is the 3' end of telomeric DNA** which is **recognised by the RNA template**, the **protein region** of the enzyme **changes conformation upon binding to the telomeric DNA**;

(Accept telomere instead of telomeric DNA)

3. Upon changing conformation, the enzyme can now **better bind to the other substrates**, which are the **free deoxyribonucleoside triphosphates** and **form bonds between the 3' end of the telomeric DNA** and the free deoxyribonucleoside triphosphates.

The Telomere Repeat Amplification Protocol (TRAP) is a Polymerase Chain Reaction (PCR) based assay that is used to determine the activity of telomerases in cell lines quantitatively by measuring the amount of telomerase products in a given time. A graph of one such assay using HeLa cells (a type of cervical cancer cell line) as positive control is shown in Fig. 2.2.

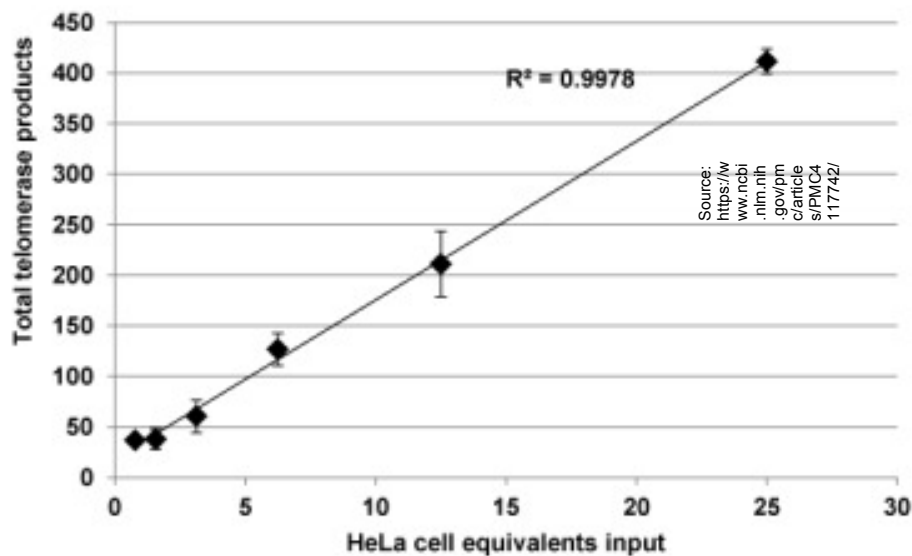


Fig. 2.2

- (b) With reference to Fig. 2.2, describe and explain the relationship between total amount of telomerase products and HeLa cell count input.

ANS [L2] (H2 Novel)

[4]

SC:	With reference to fig	describe	explain relationship
	Info from fig	DRUM with details	trends
OR:	X-axis = total HeLa cell count	X increase 1 to 25	linear / directly proportionate
	Y-axis = total telomerase products	Y increase 40 to 410	more cells = more enzymes = more ESCs

1. As the **number of HeLa cell count increases from 1 to 25**, the **total amount of telomerase products increases from 40 to 410**;
2. **linearly / directly proportionately**
3. This is because **HeLa cells**, as cancerous cells, **contain active telomerases**;
4. as the number of cell count increases, the number of telomerases increases and **more enzyme substrate complex can be formed per unit time**, resulting in more total telomerase products formed;
5. assuming there is **unlimited amount of substrate** as this is a PCR assay;

[Max 4]

Using the same technique, two other separate negative controls were set up where the results are shown in Fig. 2.3. One involved adding RNase into the mixture while the other involved heating up the HeLa cells.

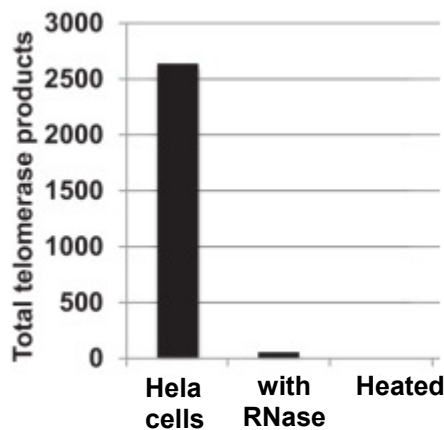


Fig. 2.3

- (c) When the cells were heated, there was no results. With reference to its structure in **Fig. 2.1**, explain why there was also very little to no result for when RNase was added to the cells culture.

ANS [L2] (H2 Novel)

[1]

SC: cells heated, no results
denatured telomerase

with reference to structure
structure shows RNA template
with exposed 3' end
RNase digests said structure

explain why no results for RNase added
give reasons

OR:

Telomerase no longer recognise / bind telomere

1. This is because **RNase digests / break down the RNA template** at the exposed 3' end of the telomerase, hence the telomerase undergoes a conformation change and **is no longer able to recognise / bind to its substrate**.

- (d) Some of the more common treatments for cancer include different drugs that inhibit telomerases.

- (i) Explain why drugs such as telomerase inhibitors on their own are often ineffective in curbing cancer.

ANS [L2] (H2 Novel)

[2]

SC: explain why
give reasons

inhibitors on their own ineffective
what does inhibitor not cure?

OR: multistep process

requires treatment for other causes of cancer

1. This is because cancer is a **multi-step process** that involves more than just active telomerases,
2. and treatment is **required** for the **many other steps** such as (list at least 1)
 - i) the gain of function mutation of proto-oncogenes
 - ii) loss of function mutation of tumour suppressor genes
 - iii) mutations causing angiogenesis
 - iv) mutations causing loss of cell adhesion

- (ii) Cancer patients sometimes develop resistance to their treatment, especially when there are tumours in different areas. Suggest why cancer cells of different tumours may exhibit different levels of resistance to the same drug.

ANS [L2/3] (H2 Novel)

SC: tumours in different areas

suggest why
give some reasons

[2]
exhibit different levels of resistance to SAME drug
why same drug different effectiveness
new functions of different mutations to lower effectiveness of drug / build resistance

OR: different mutations / causes

different variations of cells

1. **Different tumour** cells exhibit **variation**; due to (at least one) **random mutation** to certain genes / **differential gene expression** / **gene amplification**.
2. Leading to formation of new alleles coding for / more functional proteins that serve as
 - i. **transporter proteins** that **pump out** drugs.
 - ii. **Tubulin** with a non-complementary **binding site** where the drug usually binds to.
 - iii. **enzymes** that **hydrolyses** the drug.
 - iv. **inhibitors** that bind to the drugs and render them ineffective.
 AVP, e.g. cell surface receptors, proteins that regulate the cell cycle, etc

[Total: 11]

- 3 Norepinephrine, also commonly known as noradrenaline, is a hormone released during stressful or dangerous situations eliciting fight-or-flight responses. The hormone binds to protein receptors on the cell membrane to exert its effects. Fig. 3.1 shows the molecular structure of norepinephrine.

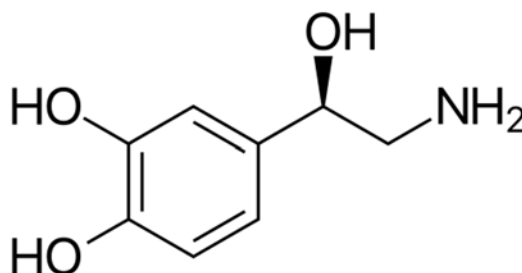


Fig. 3.1

- (a) With reference to Fig 3.1, explain why norepinephrine requires a protein receptor to exert its effects.

ANS [L2] (H2 Novel)

SC: with ref to fig
4 polar groups

explain why
give reasons

require protein receptor

OR: hydrophilic

unable to pass through hydrophobic core of cell surface membrane

[2]

1. The molecule has 4 polar groups that makes it **hydrophilic**,
2. hence it is **unable to pass through** the **hydrophobic core** of the **cell surface / plasma membrane**;
and needs to bind to a protein that is exposed to the extracellular region.

- (b) Figure 3.2 shows an example of the cell signaling pathway taken by norepinephrine in eliciting cellular responses.

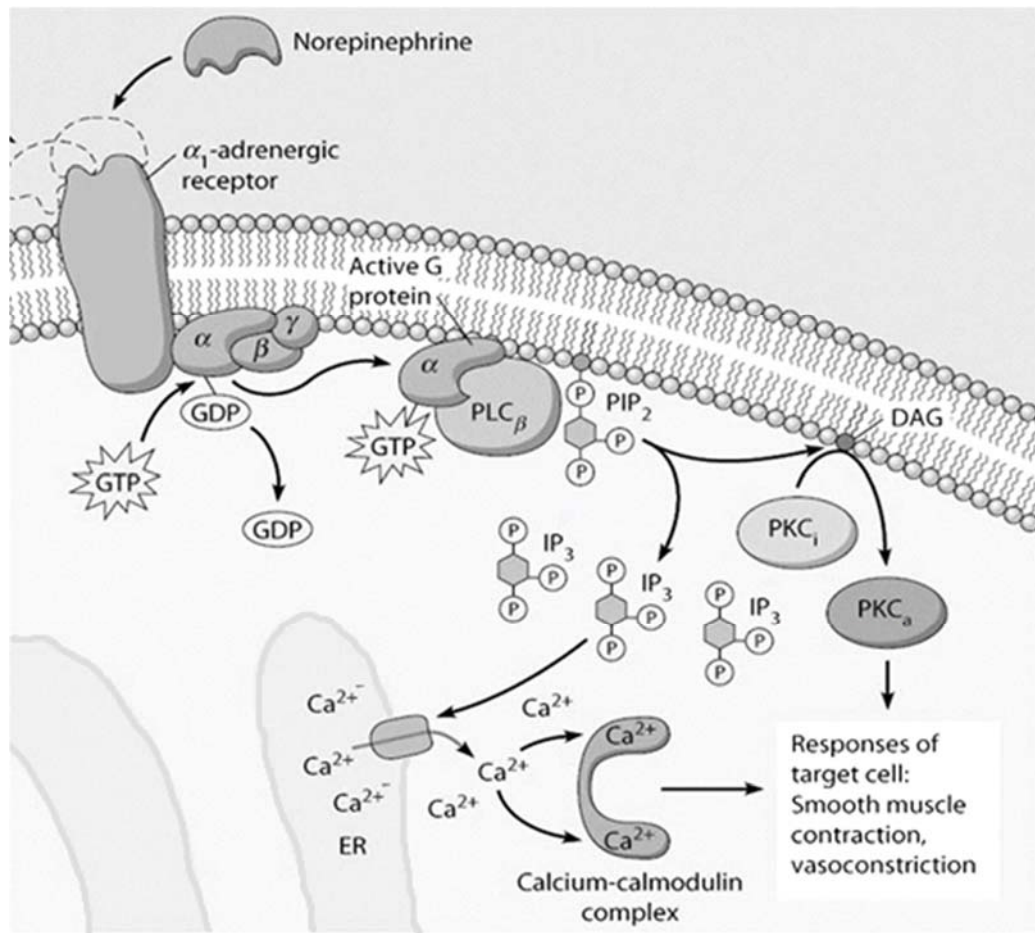


Fig. 3.2

- (i) Identify the type of receptor that norepinephrine binds to.

ANS [L2] (H2 NYJC/2016/P2/Q4 Modified)

[1]

SC: identify type of receptor
OR: give name GPCR

1. G-protein linked receptor / G-protein coupled receptor;

- (ii) With reference to Fig. 3.2, describe how an increased level of norepinephrine lead to the opening of IP₃-gated Ca²⁺ channels.

ANS [L2] (H2 NYJC/2016/P2/Q4 Modified)

[4]

SC: with ref describe how increased norepinephrine opening of IP₃ gated Ca²⁺ channels

Use info from fig process, stage by stage

- OR:
- 1] ligand binding to receptor protein (GPCR) and activate α subunit by allowing a GTP to bind
 - 2] active G protein binds to and activate PLC β which converts PIP₂ to IP₃
 - 3] IP₃ is released to Ca²⁺ ion channel
 - 4] channel undergoes conformation change and opens

1. Norepinephrine **binds to the GPCR** and **activates the α subunit by allowing a GTP to bind**,
2. Active G-protein (α subunit) **binds to and activate PLC β which converts PIP₂ to IP₃**
3. **IP₃ is released from the membrane** and diffuse to the endoplasmic reticulum where it **binds to the Ca²⁺ ion channel**;
4. Ca²⁺ ion channel **undergoes a conformational change** and then **opens**, releasing Ca²⁺ ions into the cytosol;

- (iii) With reference to **Fig. 3.2**, explain the significance of IP_3 and Ca^{2+} in the cell signalling pathway.

ANS [L2] (H2 NYJC/2016/P2/Q4 Modified)

[2]

SC: with ref to fig explain significance in cell signalling pathway
 use info: multiple activated role / function
 OR: 1 result in many secondary messengers signal amplification

- Both IP_3 and Ca^{2+} act as **secondary messengers, amplifying the signal** from one molecule of norepinephrine;

And either 2 or 3

- Each molecule of norepinephrine results in many IP_3 ;
- and each IP_3 opening the Ca^{2+} ion channel leads to many more Ca^{2+} being released;

Norepinephrine is produced during highly stressful fight-or-flight responses when an animal encounters a threat. It causes a number of effects such as increasing blood flow to the muscles and increasing alertness of the animal.

- (c) With reference to regulating stress levels in the body, explain why the production of norepinephrine by the body is also considered a form of homeostasis.

ANS [L3] (H2 Novel)

[2]

SC: produced during highly stressful encounters threat effects: increase blood flow regulate stress why homeostasis
 threat ● increase stress increase alertness
 OR: hormone counters threat by allowing body to take action bring stress down back to normal level decreases stress back to normal level

- Increasing blood flow and alertness **prepares the body to take action**, for example to **fight the threat or to run away from it**;
- By counteracting the threat, these effects help **restore stress level back to the normal level / relieve stress for the animal until the normal level**; OWTTE

(Ref. to lowering stress level to normal)

[Total: 11]

- 4 Fig. 4.1 shows the potential application of Induced pluripotent stem cells (IPSCs). Stem cells, such as those from embryos, have been increasingly used as a solution to many diseases (e.g. diabetes) though with some resistance on ethical grounds.

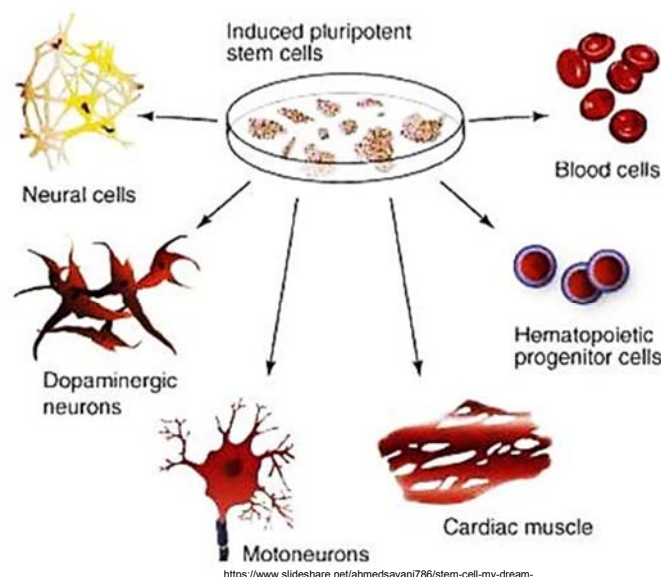


Fig. 4.1

(a)(i) State and explain what ethical issues or considerations iPSCs overcome.

ANS [L1] (Novel)

[2]

SC: State ethical implication ipsc overcome
OR: 1) source of stem cells [human embryo] morally wrong 2) ipsc from indiv own tissue

1. **Source** of most embryonic stem cells have been derived from **human embryos** that were produced from in vitro fertilization and were subsequently not used. Some feel this is **morally wrong**

or

2. **Source** of most embryonic germ cells are obtained from **aborted fetuses**. Some feel this is **morally wrong**;

Max 1

3. iPSC avoid this issue as **most do not have objections** to the **source** taken from **bone marrow transplantation**;

4. AVP

Reject Religious issues / or medical issues [tissue rejection]

(a)(ii) With reference to Fig. 4.1, explain how the ability for iPSCs to proliferate is essential in the application of iPSC.

ANS [L2] (Novel)

[1]

SC: Ref 4.1 ability IPSC proliferate essential application
OR: 1) mitotically divide to maintain stem cell number continually
2) Accept : Cell one daughter undergoes differentiation / other remain unspecialised

1. iPSCs are able to divide mitotically allowing **maintenance of stem cell number** ;

or

2. iPSC are able to divide mitotically where **one daughter cell undergoes differentiation and the other remains unspecialized**. OWTTE

MAX 1

(b) With reference to Fig. 4.1, explain how the use of iPSCs overcome immunological challenges normally associated with previous stem cell therapies.

ANS [L2] (Novel)

[2]

SC: Ref 4.1 ability IPSC overcome immunological challenges
OR: 1) own cells – convert to iPSC 2) no tissue rejection / seen with SC use previously

1. iPSCs are derived from **somatic cells / differentiated cells** taken **from the patient** needing treatment ;

2. iPSC are therefore immunologically **similar to the body** and do **not undergo tissue rejection** when reintroduced in Stem Cell Therapy. Other previous Stem cell therapies being derived from SC from other healthy individuals undergo tissue rejection and need to be introduced with immunosuppressant measures.

OWTTE

3. Include explanations suggesting the use of iPSC derived from human cells not from the patient.

4. Mention of varying degrees of rejection and how these can be overcome.

(c) With reference to Fig. 4.1, iPSC divide mitotically, explain which steps within the mitotic process ensure that iPSCs maintain their pluripotent capabilities.

ANS [L2] (Novel)

[2]

SC: Ref 4.1 IPSC divide mitotically maintain pluripotent capabilities.
OR: P M A T 1) Metaphase and Anaphase
2) ensure exact equal transfer of genome to daughter cell

1. **Metaphase** and **anaphase** of mitosis ;

2. Ensure **even / equal** transfer of the **genome** that allows each iPSC to specialize accordingly. OWTTE. Reject transfer of genetic material.

(d)(i) In the conversion of differentiated cells to iPSC, it was discovered that several changes were needed at the chromatin level as part of the reprogramming into iPSCs.

Identify one such change and explain why it is needed.

ANS [L2][High] (Novel)

[2]

SC: Diff cells to iPSC changes chromatin level reprogramming to iPSC
OR: gene silencing to activation 1] Histone acetylation 2] neutralise histone tails – unwind chromatin
Accept 3] Histone demethylation 4] free histones – toward a more linear dna

1. **Histone acetylation** - Which causes the addition of acetyl groups to histone tails, neutralizing them allowing chromatin to unwind as linear dna.
2. linear dna allows gene activation and direct transcription of normally silenced genes
3. **Histone demethylation**-Which causes the **removal of methyl groups** allowing **histones to be less compact**;
4. one step closer toward linear dna, allows gene activation and direct transcription of normally silenced genes

OWTTE.Max 2

(d)(ii) In relation to (d)(i), state one other modifications would be needed at the linear dna level.

ANS [L2] (Novel)

[1]

SC: (d)(i) other modifications linear dna level
OR: gene silencing to activation 1] dna demethylation

1. **DNA demethylation** ;

[Total: 10]

- 5 Fig. 5.1 shows the activation of transcription initiation in eukaryotes by recruitment of the eukaryotic RNA polymerase II holoenzyme complex

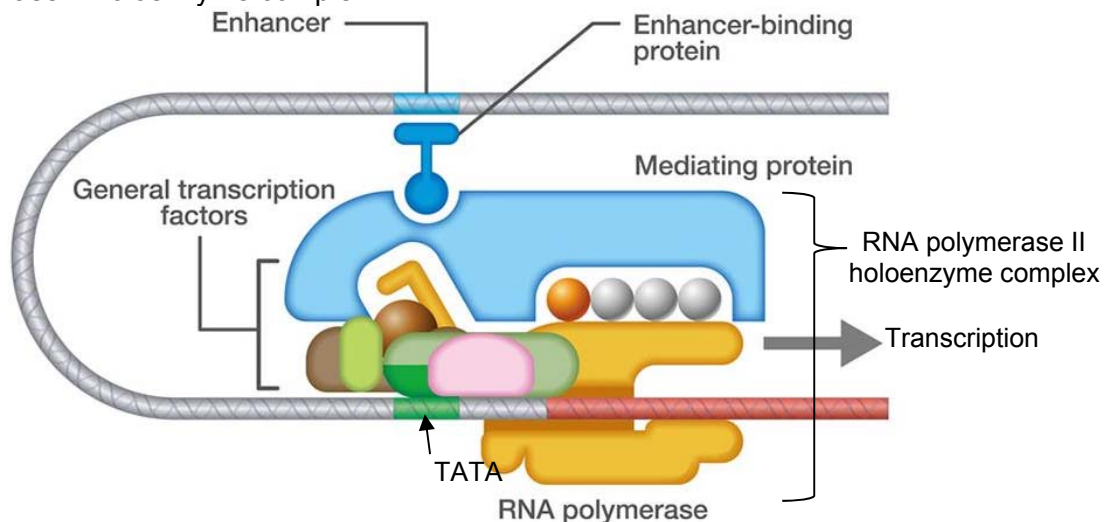


Fig. 5.1

- (a) With reference to Fig.5.1, identify a way in which transcription in eukaryotes differs from that in prokaryotes.

ANS [L2] (modified H2 NJC/2009/P2/Q3)

[1]

SC: with reference to Fig.5.1 identify 1 way in transcription eukaryotes vs. prokaryotes
OR: visible in fig. 5.1 difference only 1. TATA vs. Pribnow box
2. RNA pol II holoenzyme complex + GTF to promoter RNA pol to promoter

1. In eukaryotes, transcription begins when **general transcription factors** and **RNA polymerase II holoenzyme complex** binds to promoter whereas in prokaryotes, transcription begins when **RNA polymerase** binds to promoter;
2. In eukaryotes, RNA polymerase II holoenzyme complex (and General transcription factors) binds to the **TATA box** while in prokaryotes, RNA polymerase binds to **Pribnow box**;
3. AVP;[Max 1]

- (b) Eukaryotic DNA contains both exons and introns. Fig. 5.2 shows the structure of a length of pre-mRNA formed from one gene.

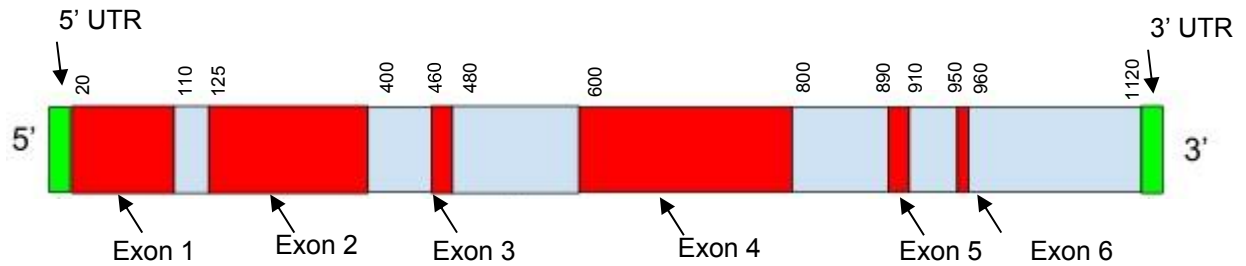


Fig. 5.2

- (i) Calculate the longest mRNA molecule that can be formed from the pre-mRNA shown in Fig. 5.2

ANS [L2] (modified H1 Alevel/2013/P2/Q1)

[1]

SC: Calculate longest mRNA

OR: 20-110; 125-400, 400-460, 460-480, 600-800, 890-910, 950-960

Longest mRNA include Exons 1-6

Length of longest mRNA molecule = $90 + 275 + 60 + 20 + 200 + 20 + 10 = 675$

- (ii) Outline how this gene, shown in Fig. 5.2, could code for a number of different proteins.

ANS [L2] (modified H1 Alevel/2013/P2/Q1)

[1]

SC: Outline how could code for different protein

OR: Alternative RNA splicing;

1. **Alternative splicing** with different combinations of exons could be spliced together to form mRNA which code different proteins
2. AVP;

- (c) Eukaryotes and prokaryotes contain ribosomes, an integral cell structure responsible in protein synthesis. These ribosomes are made up of ribosomal RNA (rRNA) which is single stranded and unbranched polynucleotide chain. However, 70% of it is in the form of double stranded and helical. These double stranded regions form "hairpin loops", shown in Fig. 5.3.

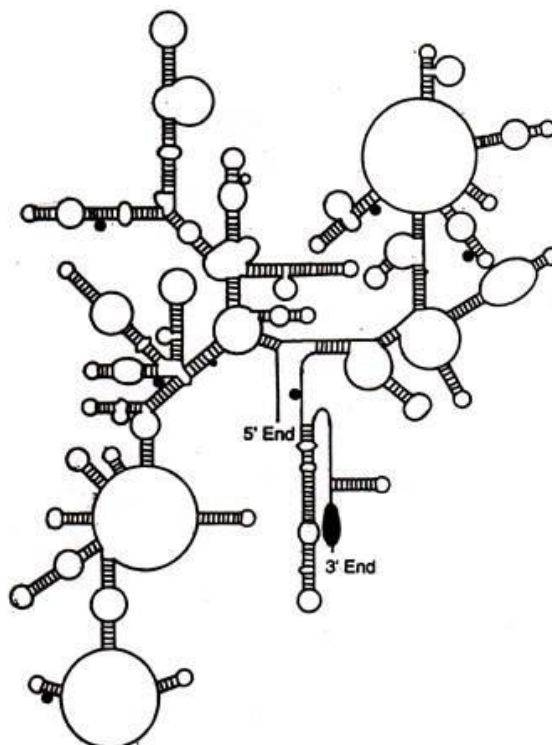


Fig. 5.3

With reference to Fig. 5.3, suggest why some parts of the molecule are double stranded.

ANS [L3] Novel

SC: With reference to Fig. 5.3
OR:

suggest why
reasons (x 2)

molecule (ds)
complementary base pairing;
high number of H bonds (remain ds)

[2]

1. There are **different sequences** that allow **complementary base pairing**.
2. There is a **high number of hydrogen bonds** allowing it to remain double stranded.

Fig. 5.4 shows a structure of the tryptophan (*trp*) operon of *Escherichia coli*, a prokaryote.

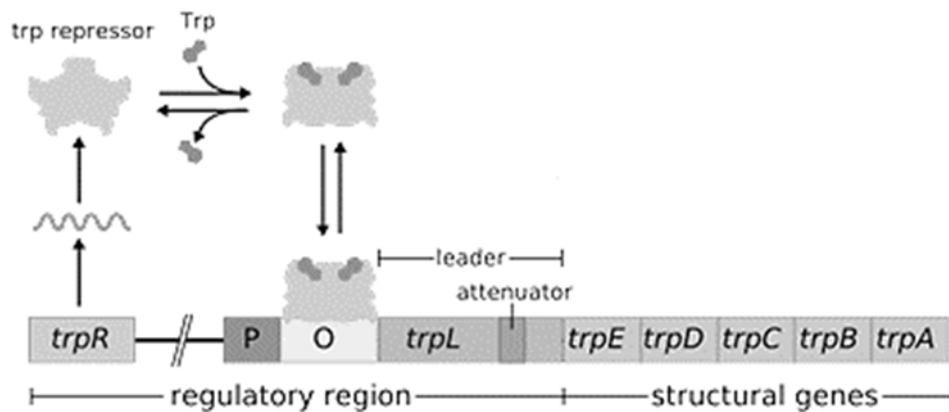


Fig. 5.4

(d) With reference to Fig. 5.4, explain the role of an operator in *trp* operon.

ANS [L1] Novel

SC: With ref. Fig. 5.4
OR:

explain role of operator
regulate multiple genes

trp operon
bind to trp repressor protein

[1]

1. The operator regulates the **multiple genes** downstream (i.e. *trpE* – *trpA*) and is the site where the **trp repressor protein binds**.

(e) Explain the effect on the transcription of the operon if trp repressor protein is mutated.

ANS [L2] (H2 RI Prelim/2011/P2/Q3)

SC: Explain
OR:

effect transcription

trp repressor mutated
mutated trp repressor

not able to bind to corepressor, tryptophan;
no binding to operator; RNA pol always bound to promoter
transcription is always happen

[2]

1. Trp repressor protein, encoded by the regulatory gene, **not able to bind to tryptophan a.a. / corepressor / OWTTE**
2. Trp repressor not bound to operator region, allowing RNA polymerase to **always bound to promoter**, leading to transcription of structural gene is **always on / OWTTE**

- (f) Explain the significance of negative feedback in both inducible and repressible operons in prokaryotes.

ANS [L2] Novel

SC: Explain
OR:

the significance
importance? negative feedback

both inducible & repressible operons
regulate gene expression
synthesise enzymes only when needed
repressed enzyme production if product/substrate is available
save overall cell's resources and energy

[2]

1. It allows bacteria to regulate gene expression according to the changes in the environment
2. It allows synthesis of enzymes only when needed
3. It allows repression of enzymes only when the products/substrates are available
4. It allows cells to save overall energy and resources and not to synthesise enzymes/proteins when it is not needed. / OWTTE

[Any 2]

[Total: 10]

- 6** The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced in cells called melanocytes by the action of the enzyme tyrosinase. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH), a hormone that is normally present in healthy cattle, binds to a receptor coded for by the gene, **R**. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for an inhibitor protein that binds to and blocks the MSH receptors coded for by **R**, thus preventing stimulation of tyrosinase activity in a melanocyte. The recessive allele, **b**, does not produce a functional protein.

- (a) (i) State the name given to the interaction between the **R** and **B** gene loci.

ANS [L1] (H2 ACJC/2017/P2/Q4)

SC: state name interaction
OR: epistasis; dominant suppression epistasis

[1]

1. Epistasis; Dominant suppression epistasis

- (ii) Identify all possible genotype(s) for the black coat phenotype for the Norwegian cattle and explain how the alleles interact to give rise to the black colouration.

ANS [L2] (H2 ACJC/2017/P2/Q4 modified)

SC: ID all possible genotypes black phenotype
OR: Draw the fishbone to test hypothesis

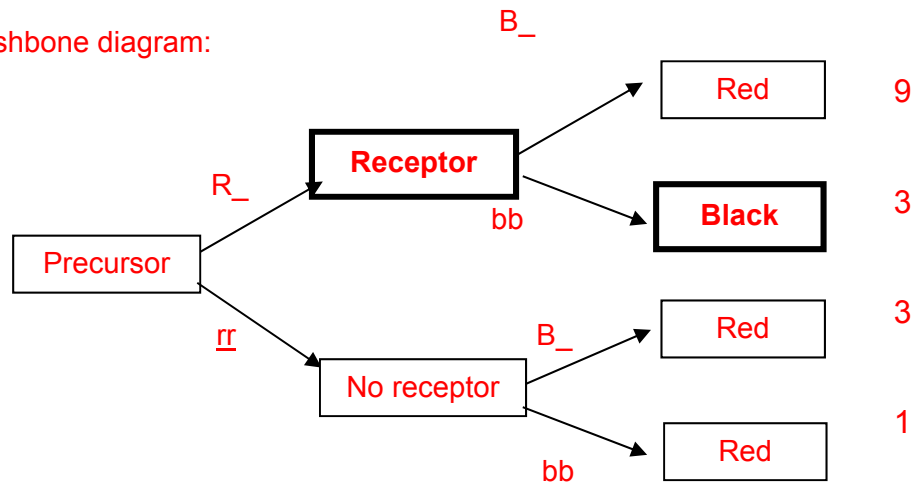
explain how alleles interact
cause and effect give expression of genotype
only when receptor present and inhibitor absent
To allow MSH to bind and increase tyrosine activity

[2]

Hence, only **R_bb** which are **Rrbb** and **RRbb**

1. **Rrbb** and **RRbb**
2. Only for genotypes where both the **receptor is present** and the **inhibitor absent**, will the **MSH bind to the receptor** and **increase the activity of tyrosinase** to give rise to a black phenotype.

Fishbone diagram:



- (b) A red cow, with genotype **RRBB** is mated with a bull which is homozygous recessive at both gene loci. Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the F₁ and F₂ generations.

ANS [L2] (H2 ACJC/2017/P2/Q4)

[5]

SC: **RRBB X homozygous recessive at both gene loci**
RRBB X rrbb

OR:

Parent phenotypes	Red bull	x	Red cow	
Parent genotypes	<u>rrbb</u>		RRBB	1m
Gametes	<u>rb</u>		RB	
F ₁ genotype	RrBb			
F ₁ phenotype	All red			1m
F ₁ cross	RrBb	x	RrBb	
F ₁ gametes	RB	Rb	rB rb	1m

Punnett Square:

	RB	Rb	rB	rb	
RB	RRBB Red	RRBb Red	RrBb Red	RrBb Red	
Rb	RRBb Red	RRbb Black	RrBb Red	Rrbb Black	
rB	RrBB Red	RrBb Red	rrBB Red	rrBb Red	
rb	RrBb Red	Rrbb Black	rrBb Red	rrbb Red	1m

F₂ phenotypic ratio 13 red coat : 3 black coat 1m

Mark allocation:

1. Correct parent gametes;
2. Correct F₁ phenotype and genotype;
3. Correct F₁ gametes;
4. Correct F₂ phenotypes and genotypes in Punnett square;
5. Correct F₂ phenotypic ratio;

[5]

[Total: 8]

- 7 Fig. 7.1 shows a flowchart showing the process of cellular respiration in a muscle cell.

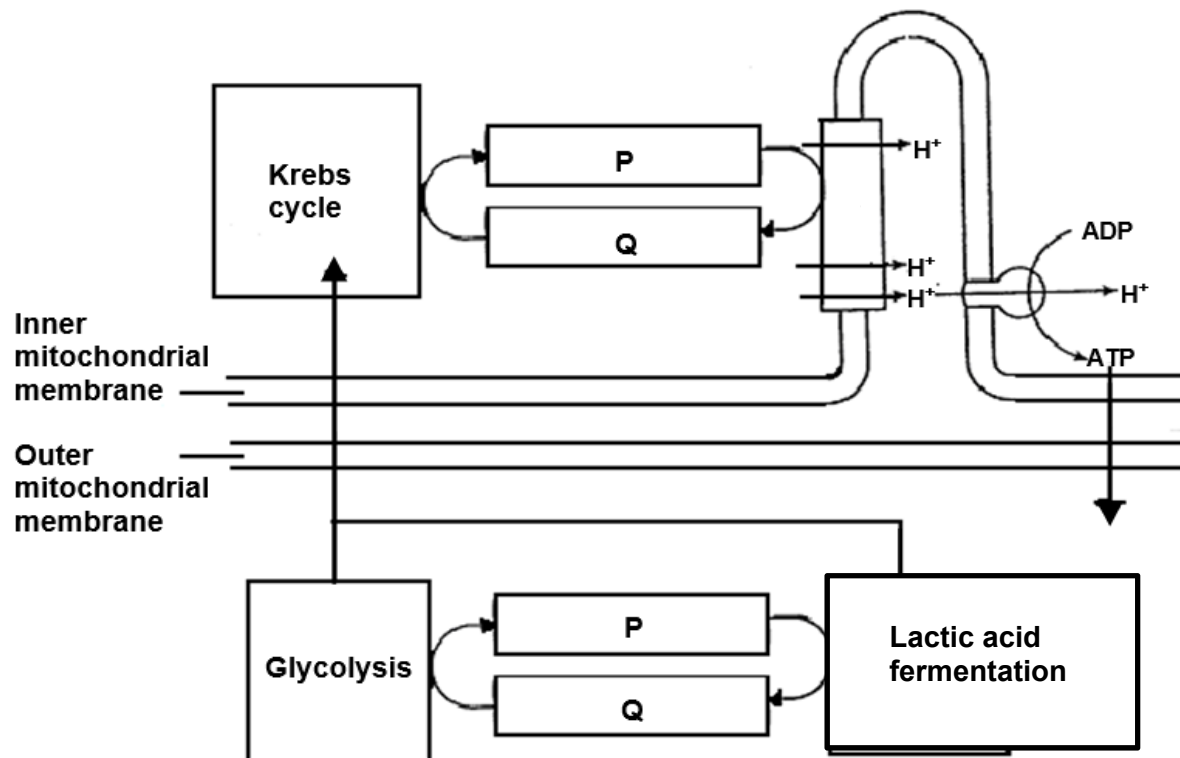


Fig. 7.1

- (a) P and Q plays an important role in ensuring that glycolysis always happen in either aerobic or anaerobic respiration process.
- (i) Identify P and Q.

P:

Q:

[1]

ANS [L1] (H2 HCI/2009/P2/Q2)

[1]

SC: identify P Q
OR: NAD⁺ NADH

P: NADH / reduced NADH

Q: NAD⁺

(ii) Explain the significance of P and Q in linking Krebs cycle to the formation of ATP in the cristae.

ANS [L3] (H2 HCI/2009/P2/Q2)

[2]

SC: Explain significance of P and Q link Krebs cycle to ATP formation
OR:

1. **Q / NAD⁺** receives electrons **from the respiratory intermediates / substrates in the Krebs cycle** which completes glucose oxidation and is **reduced to P / NADH** which **carries** high energy **electrons to electron transport chain (ETC)** at the cristae / inner membrane of mitochondria;
2. **Energy released** during the passage of electrons down ETC is coupled to **formation of proton gradient** which will be harnessed by **ATPase** to power **chemiosmosis / OWTTE**;

(b) With reference to Fig. 7.1, explain why lactic acid fermentation occurs.

ANS [L2] (H2 HCI/2009/P2/Q2)

[2]

SC: With ref. Fig. 7.1 explain why lactic acid occurs
OR:

1. There is a **lack of / no oxygen** present and muscle cell switches to anaerobic respiration to **regenerate NAD⁺** from NADH so that glycolysis can still be ongoing / OWTTE
2. where pyruvate is reduced/converted to lactate for continual **ATP production** by substrate level phosphorylation.

(c) Pyruvate from glycolysis has to be converted to acetyl CoA in a link reaction.

State the location where link reaction occurs.

ANS [L1] Novel

[1]

SC: State location link reaction occurs
OR:

1. **Mitochondrial matrix**

(d) Fig. 7.2 shows the Krebs cycle process.

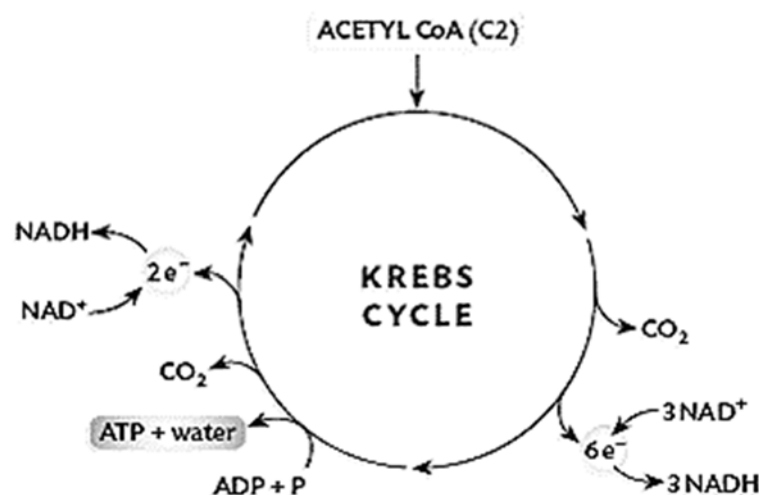


Fig. 7.2

Suggest a significance of acetyl-CoA in metabolism of *other* respiratory substrates.

ANS [L3] Novel

[2]

SC: Suggest significance of acetyl-CoA metabolism of other respiratory substrate
OR: important in lipid metabolism; fatty acids are broken down to acetyl CoA ; increase energy pdtion

1. Acetyl CoA is an important product from **lipid metabolism**

2. where fatty acids are broken down to acetyl CoA in order to **increase energy production**
- (e) In plant cells, on the other hand, has organelles where their cellular respiration synergizes with photosynthesis process. Calvin cycle, a light independent reaction, occurs in the stroma of chloroplast which plays an important role in photosynthesis.

Outline 2 ways in which Calvin cycle differs from Krebs cycle.

ANS [L2] (H2 Alevel/2008/P2/Q8)

SC: Outline 2 ways

differs

OR: differences between Calvin cycle and Krebs cycle

[2]

	Features	Calvin cycle	Krebs cycle
1	Location	Occurs at the stroma of chloroplast	Occurs at the mitochondrial matrix
2	Starting material	Ribulose biphosphate (RuBP) is the starting material of the cycle and eventually regenerated	Oxaloacetate is the starting material of the cycle and eventually regenerated
3	Substrate	Carbon dioxide is used as the substrate	Acetyl-CoA is used as the substrate
4	Role of CO ₂	Carbon dioxide is needed for carbon fixation. It converts RuBP to form unstable compound 6C which eventually breaks down to glycerate-3-phosphate (GP)	Carbon dioxide is released as a result of decarboxylation reactions
5	Electron carriers	NADPH is needed to reduce glycerate-3-phosphate (GP) to triose phosphate (TP) / glyceraldehyde-3-phosphate by serving as electron donors	NAD⁺ and FAD are required for oxidation of intermediates of the cycle by serving as electron acceptors
6	ATP	Needed energy through hydrolysis of ATP	ATP synthesis through substrate level phosphorylation
7	Products	For every 3 molecules of CO₂ , one GP is made	Each cycle gives rise to 2 ATP, 6 NADH, 2 FADH₂
8	Process type	Anabolic due to synthesis of hexose sugar phosphates and eventually energy storage polysaccharide	Catabolic due to oxidation of Acetyl-CoA through series of decarboxylation and dehydrogenation

9. AVP;

[Max 2]

[Total: 10]

- 8 Fig. 8.1 shows the distribution of the salamander *Ensatina* of the Pacific Coast region of the United States. Their collective range stretches approximately 1400km down from the Canadian coast, forking into a 'coastal arm' and an 'inland arm' at the dry, hot lowland region of the Central Valley; till the ranges finally meet again upon reaching the Southern Californian Coast. The first salamanders were hypothesized to have originated in the north.

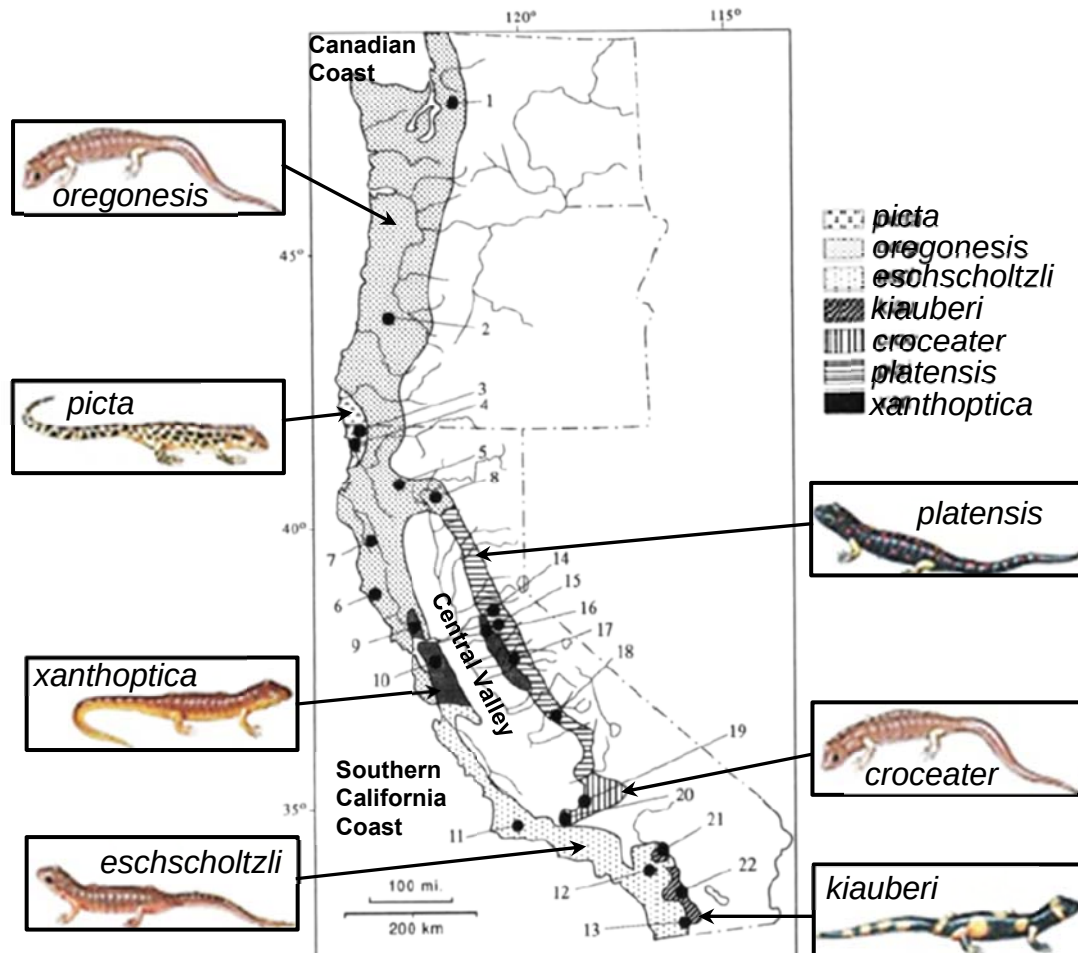


Fig. 8.1 Range of the salamander *Ensatina* of the Pacific Coast region of the United States. Numbered labels represent sample collection sites. <http://sysbio.oxfordjournals.org/> at North Carolina State University on March 25, 2014

- (a) Explain the role of natural selection in evolution.

ANS [L1] (Novel)

SC: Role Natural Selection

OR: 1a] Adv allele selected for

Evolution

1b] survive the selection pressure and reproduce

1. Natural Selection is the mechanism in which an **advantageous allele / gene / trait** is **selected for-**

Allowing individuals within the population to survive / prove as an advantage under the selection pressure and reproduce;

OWTTE

[1]

E. picta, found exclusively on the eastern coast of California has a range of less than a hundred kilometers compared to its ancestor *E. oregonesis* which extends to an excess of a thousand kilometers.

- (b) With reference to Fig. 8.1 suggest why the range of *E. picta* is so limited compared to *E. oregonesis*.

ANS [L3] (Novel)

[4]

- SC: ref 8.1 suggest why ranges *E. picta* limited *E. oregonesis*
 OR: 1] different selection pressures Possible : Predation
 2] Advantageous trait : camouflage [spotty] adapted to specific type of vegetation
 3] limited range- dependent on vegetation range, otherwise conspicuous
1. *E. picta* **spotty morphology** suggests **camouflage** adapted to a type of vegetation specific to the region [advantageous trait];
 2. The trait allowed a selective advantage perhaps against predators or advantageous during foraging.
 3. The range of *E. picta* was however **limited to the range of the specific vegetation** it was adapted to, which may be limited to that region.
 4. The same spotty morphology may make *E. picta* **conspicuous to predators outside** the range of that **specific vegetation**.
 OWTTE

It was later discovered that *E. oregonesis*, *E. picta* and *E. xanthoptica* were found to have fertile hybrids within the population; while further south *E. eschscholtzii*, *E. platensis*, *E. croceator* and *E. kiauberi* were incapable of interbreeding.

- (c)(i) Discuss what conclusions can be derived from the mating study of the seven groups of Salamander.

ANS [L2] (Novel)

[2]

- SC: conclusions mating study 7 groups Salamander
 OR: 1] hybrids in population *E. oregonesis* / *picta* / *xanthoptica* sub species
 2] no interbreeding *E. eschscholtzii*, *E. platensis*, *E. croceator* and *E. kiauberi* speciation taken place
1. Presence of **fertile hybrids in population** indicate that *E. oregonesis*, *E. picta* and *E. xanthoptica* were **subspecies**. / OWTTE.
 2. That *E. eschscholtzii*, *E. platensis*, *E. croceator* and *E. kiauberi* were **unable to interbreed** indicated that **speciation** may have taken place

- (c)(ii) Explain the application of a known species concept upon which your answer in (c)(i) is based upon.

ANS [L1] (Novel)

[2]

- SC: application known species concept which (c)(i) based upon
 OR: 1] Biological spp Reproductive isolation
1. **Biological Species Concept** ;
 2. Based on presence or absence of **reproductive isolation** ;
 OWTTE

- (d)(ii) What reliable technical method can be used to better support your answer in (c)(i)?

ANS [L2] (Novel)

[1]

- SC: Reliable Technical method support (d)(i)
 OR: 1] Molecular homology [MtDNA/ mRNA/ protein etc]
 2] Quantitative, comparable across most living things
1. **Molecular homology** use of [MtDNA/ mRNA/ protein etc];
 AVP / OWTTE

[Total: 10]

Notes: Ring Species: Salamanders: http://www.pbs.org/wgbh/evolution/library/05/2/l_052_05.html

Some critics of the theory of evolution argue that it doesn't convincingly explain the origin of new species. They say that members of one species couldn't become so different from other individuals through natural variation that they would become two separate non-interbreeding species.

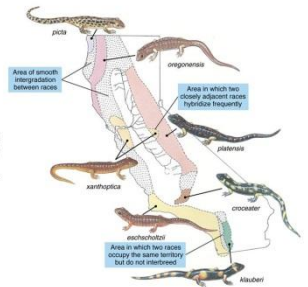
One of the most powerful counters to that argument is the rare but fascinating phenomenon known as "ring species." This occurs when a single species becomes geographically distributed in a circular pattern over a large area. Immediately adjacent or neighboring populations of the species vary slightly but can interbreed. But at the extremes of the distribution -- the opposite ends of the pattern that link to form a circle -- natural variation has produced so much difference between the populations that they function as though they were two separate, non-interbreeding species.

In concept, this can be likened to a spiral-shaped parking garage. A driver notices only a gentle rise as he ascends the spiral, but after making one complete circle, he finds himself an entire floor above where he started.

Ring species—salamanders

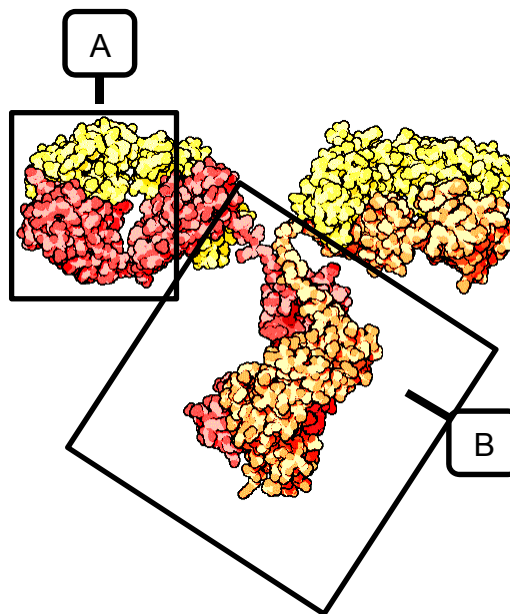
The ensatina salamander (*Ensatina eschscholtzii*) occurs from Canada to Southern California with interbreeding between adjacent populations through this range.

The Central Valley—a dry, hot lowland area—is divided into a coastal arm and inland arm. However, where these two arms of the species meet again in Southern California, interbreeding does not occur.



- 9 Fig. 9.1 shows the globular structure of a B cell receptor IgM after the B cell has just left the lymph node.

Fig. 9.1



- (a) State the developmental stage of these B cells as part of the cellular innate defense.

ANS [L1] (Novel)

SC: dev stage cellular innate defence
OR: 1] Mature [IgM] 2] naïve

1. **Mature** -given the presence of IgM and naïve in that no encounter with the antigen has taken place;

- (b) With reference to Fig. 9.1, region A outlines a section of IgM that is involved in recognition. State and explain the process occurring in lymph node that this cell had undergone upon meeting a specific pathogen.

ANS [L2] (Novel)

SC: Ref 9.1 Region A recognition
OR: variable region antigen binding site

process Lymph node
1] somatic hypermutation

2] introduction of point mutation on variable region
3] Affinity Maturation

1. **Somatic Hypermutation ;**

Either 2 or 3

2. Introduction of **point mutations on the variable region**
3. Promoting **Affinity Maturation**; AVP, OWTTE

- (c) With reference to Fig. 9.1, explain why region B is different from corresponding regions of other similar structures like IgD, IgG from the same type of B Cells.

ANS [L2] (Novel)

[2]

SC: Ref 9.1 Region B constant diff corresponding regions IgD, IgG same type B cell
OR: 1] class switching 2] alternative gene splicing of constant region DNA

1. **Class switching**
2. **Alternative gene splicing** by spliceosomes at the **constant region**;

Max 2

- (d) Explain how immunological memory is established and its significance.

ANS [L2] (Novel)

[3]

SC: importance immunological memory is established significance
OR: 1] Memory T Cells 3] faster recognition / response
2] Memory B Cells

1. **Memory T cells differentiated from activated T Cells / [CD4+ T Cell]**
2. **Memory B cells differentiated from activated T Helper Cells and Activated B Cells.**
3. Significance – allows a **faster recognition** and **response** to the **same pathogen**

- (e) Type 1 diabetes is a chronic disease. In type 1 diabetes, beta cells (which make insulin) in the pancreas are destroyed by the body's immune system.
Suggest at which stages of the innate immune system may be the cause for Type 1 diabetes, explain your answer.

ANS [L3] (Novel)

[2]

SC: Type 1 diabetes beta cells destroyed by immune system innate system stage cause
OR: autoimmune 1] somatic recombination- different receptors /pathogens
2] Clonal selection defective / Clonal deletion impaired

1. **Somatic recombination**- within Thymus and Bone marrow generate different cell types with varied receptors against pathogens.

Either 1B or 1C

1B] Some of these cell types recognize the body's own cells and must undergo **Clonal deletion**, Type 1 diabetes some of these self-recognizing cell types are not deleted resulting in the destruction of beta cells;

or

1C] Some of these cell types recognize the body's own cells and because of **defective Clonal Selection**, these self-recognizing cell types are not deleted but are selected for, resulting in the destruction of beta cells;

Or

2. **Somatic hypermutation**

2B] Where mutations introduced inadvertently give rise to host specific antibodies.

3. **AVP**

[Total: 10]

10 Fig. 10.1
combined
land and

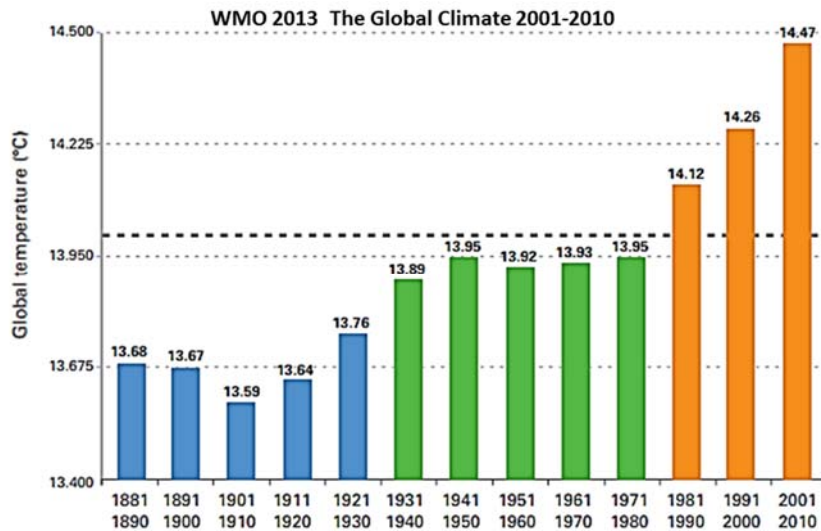


Fig. 10.1

shows the global
surface air
temperature over
sea.

(a)(i) As seen in Fig. 10.1 global temperatures have been on a constant rise, give two gases and their respective relative global warming potentials, which would contribute to this global phenomenon.

ANS [L1] (Novel)

SC: Global temp rise

OR:

two gases

1] CO₂

2] CH₄ methane

GWP

1

25

global phenomenon

Green house effect- global warming

[2]

1. Carbon dioxide gas, and Methane ;
2. Methane has a higher GWP compared to Carbon dioxide 25 to 1.

AVP;

(a)(ii) With a named example, explain what is positive feedback in global warming.

ANS [L2] (Novel)

SC: named example

OR: positive feedback loop

(+ve) impact loop global warming

1] polar ice melts

2] less reflection of solar radiation
increased rise temp-more ice melting

[2]

1. Higher global temperatures cause greater Polar Ice Melts
2. This in turn causes there to be less ice reflecting solar radiation which therefore raises global temperatures accordingly. Positive feedback loop;
3. Permafrost melting releasing methane concentration.

AVP;

(b) Global warming also has had significant effects on the ranges of both animals and plants. Explain what some of these effects are.

ANS [L2] (Novel)

SC: global warming

OR: Rise in temperature

sig effects on ranges

1] increased range toward poles latitudinal shift

4] increased range up mountains altitudinal shift

5] Altitudinal and latitudinal limitation of range still due to temp barrier

animal

2] wider and faster

plant

3] less wide- dispersal

[3]

1. Increased range of both animals and plants toward poles Latitudinal shift
2. Animal range wider and faster due to mobility and spread;
3. Plant ranges more limited due to dispersal
4. Increased range up mountains, altitudinal range
5. Altitudinal and latitudinal temp barriers still apply limiting ranges.

6. AVP

Only 22% of Canadian bird species spend the whole year in Canada (Fig. 10.2A). Most others migrate to the United States (33%), to Central America, Mexico and the Caribbean (23%) or to South America (15%). Some travel to Europe or Asia or spend long periods of time at sea (7%).

A forty year study of these migratory species (Fig. 10.2B) has shown declines in species migrating to the United States, Mexico and very drastic declines in Brazil, South America.

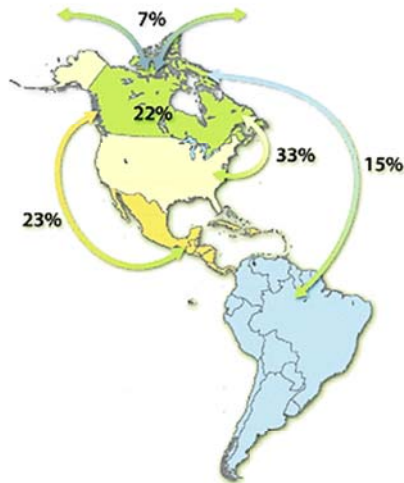


Fig.10.2A Destinations of Canada's migratory bird species

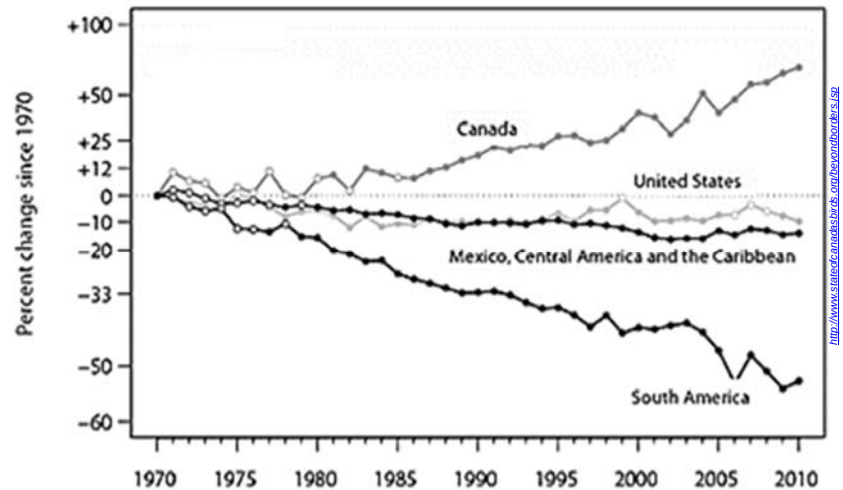


Fig.10.2B Indicators of average population status of species that primarily winter in four major

(c)(i) With reference to Fig. 10.2A&B, suggest what might be a possible direct anthropomorphic cause for the decline in Brazil, explain why.

ANS [L3] (NOVEL)

SC: decline in migratory spp

OR: Loss of numbers

direct anthropomorphic cause

1] Deforestation- loss of habitat

largest decline Brazil

[2]

1. **Deforestation** which causes the

2. **loss of habitat** in more significant over the years as deforestation progresses;

AVP;

Reject: enlarging cities / choice of food as these cause deforestation and so are indirect causes.

(c)(ii) With reference to Fig. 10.2B, suggest a biological impact of the decline of migratory species in Brazil.

ANS [L3] (NOVEL)

SC: ref 10.1B

OR: Loss of numbers

biological impact

1] Loss of Biodiversity

2] effect of ecosystem in Brazil

largest decline Brazil

[1]

1. **Loss of Biodiversity** as migrant spp have fewer places to tide the winter

2. **Secondary effects on ecosystems** in the Brazilian rainforest where migratory birds may have significant interactions and effects on both flora and fauna.;

AVP;

Max 1

[Total: 10]

